Parametrization of curves

What is a parametrization?

Example: The circle of radius 1 about $(0,0)$ has two descriptions:

- Implicit: $x^2+y^2=1$
- Parametrization:

$$\begin{cases} x = \cos t \\ y = \sin t \end{cases}$$

$0 \le t < 2\pi$ ← cover circle, exactly once

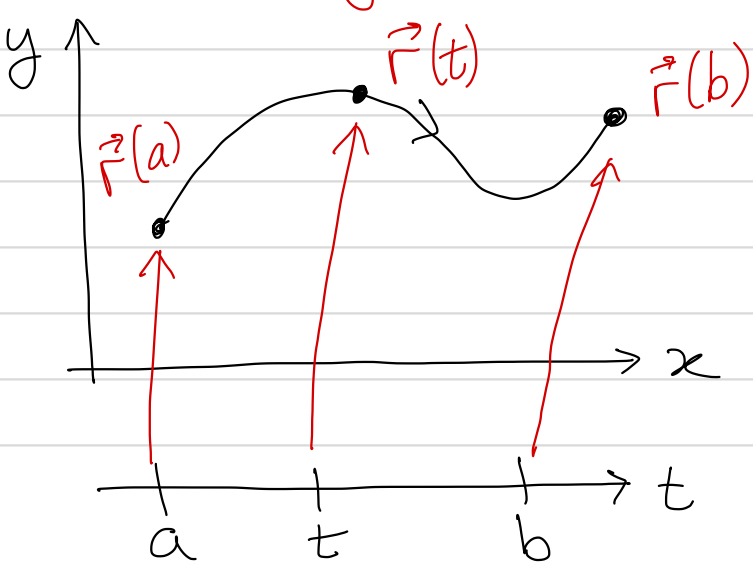
t ← parameter

Same information:

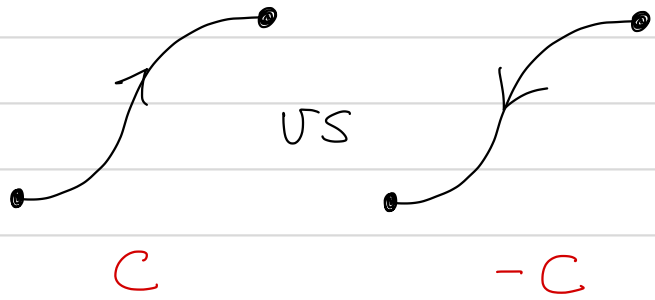
$$\vec{r}(t) = (\cos t, \sin t)$$

In general:

$$\vec{r}(t) = (x(t), y(t)), \quad a \le t \le b$$



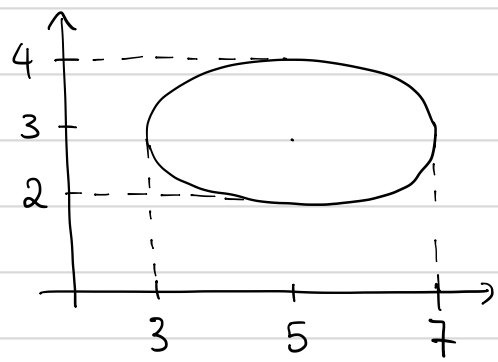
Orientations:



Examples:

1) Ellipse

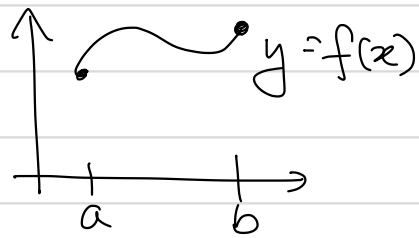
$$\frac{1}{4}(x-5)^2 + (y-3)^2 = 1$$



$$\begin{cases} x = 5 + 2\cos t \\ y = 3 + \sin t \end{cases}$$

$0 \le t \le 2\pi$

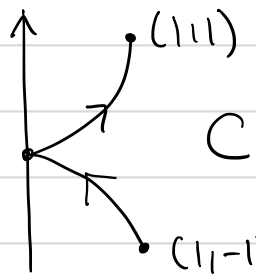
2) Graph



$$\vec{r}(x) = (x, f(x)), \quad a \le x \le b.$$

3) Cusp.

$y^2 = x^3$

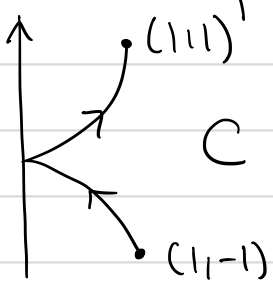


$C = C_1 \cup (-C_2)$

$C_1 : \begin{cases} x = x \\ y = +\sqrt{x^3} \end{cases}$ $(0 \leq x \leq 1)$

$C_2 : \begin{cases} x = x \\ y = -\sqrt{x^3} \end{cases}$ $(0 \leq x \leq 1)$

3') A "better" parametrization:



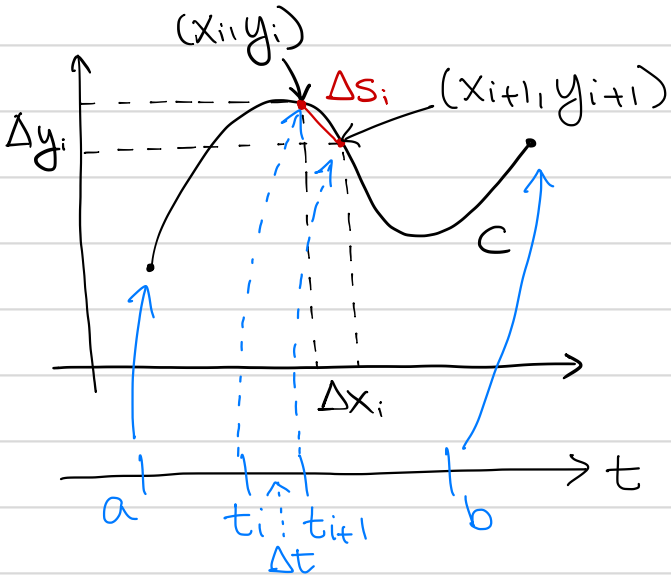
C

$\begin{cases} x = t^2 \\ y = t^3 \end{cases}$ $(-1 \leq t \leq 1)$

Line integrals (16.2)

↳ of functions

↳ of vector fields (work)



Parametrization:

$$\vec{r}(t) = (x(t), y(t)), \quad a \leq t \leq b.$$

$$\text{length}(C) = \lim_{\Delta s_i \to 0} \sum_i \Delta s_i$$

$$= \int_a^b \underbrace{\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt}_{ds \text{ "arc length element"}}$$

Why:

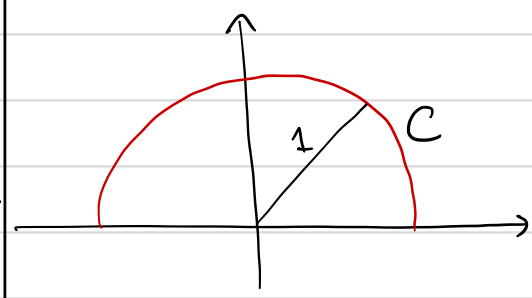
$$\Delta s_i = \sqrt{(\Delta x_i)^2 + (\Delta y_i)^2}$$

$$= \sqrt{\left(\frac{\Delta x_i}{\Delta t}\right)^2 + \left(\frac{\Delta y_i}{\Delta t}\right)^2}\ \Delta t$$

Definition:

$$\boxed{\begin{aligned} \int_C f(x,y)\, ds &= \lim_{\Delta s_i \to 0} \sum_i f(x_i, y_i)\, \Delta s_i \\ &= \int_a^b f(x(t), y(t)) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt \\ &= \int_a^b f(\vec{r}(t))\, |\vec{r}\,'(t)|\, dt \end{aligned}}$$

Example: $f(x,y) = 2 + x^2 y$



$x = \cos(t)$

$y = \sin(t)$

$0 \leq t \leq \pi$

$$\int_C f(x,y)\, ds =$$

$$= \int_0^\pi (2 + \cos^2 t \sin t) \underbrace{\sqrt{(\sin t)^2 + (-\cos t)^2}}_{1}\, dt$$

$$= 2t - \frac{\cos^3 t}{3}\Big|_0^\pi = 2\pi + \frac{2}{3}$$

Applications of $\int_C f\,ds$:

1) $\int_C ds = \text{length of } C =: L$

2) Average value of f on C $= \frac{1}{L}\int_C f\,ds$

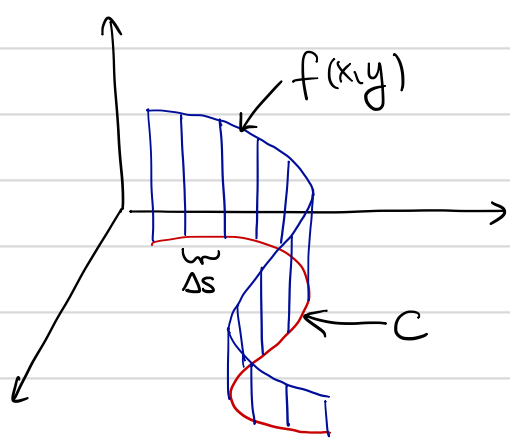
3) $\rho(x,y)$ = density of C (mass / unit length) at (x,y)

$$M = \int_C \rho\,ds$$

$$\bar{x} = \frac{1}{M}\int_C x\rho\,ds$$

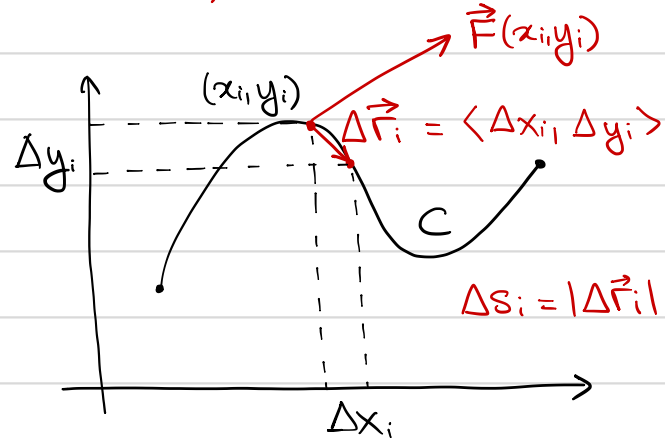
$$I_y = \frac{1}{M}\int_C (x^2+z^2)\rho\,ds$$

4) In 2D: area of "fence"



Line integrals of vector fields

$$\vec{F} = \langle P, Q, R \rangle$$



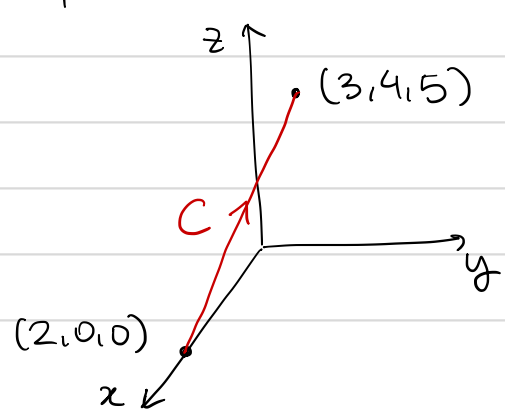
$$\vec{r}(t) = (x(t), y(t), z(t)), \quad a \le t \le b$$

Definition (Work): $\vec{F} = \langle P, Q, R \rangle$

$$\underbrace{\int_C \vec{F}\cdot d\vec{r}}_{\text{"Work"}} := \lim_{\Delta r_i \to 0} \sum_i \vec{F}(x_i,y_i)\cdot \Delta\vec{r}_i$$

$$= \int_a^b F(\vec{r}(t))\cdot \frac{d\vec{r}}{dt}\,dt$$

Example: $\vec{F} = \langle y, z, x \rangle$



$$\vec{r}(t) = (2,0,0) + t\big((3,4,5)-(2,0,0)\big)$$
$$= (2+t,\ 4t,\ 5t)$$
$$0 \le t \le 1$$

$$\int_C \overbrace{\vec{F}}^{\langle y,z,x\rangle} \cdot \overbrace{d\vec{r}}^{\langle dx,dy,dz\rangle} =$$

$$= \int_0^1 \langle 4t, 5t, 2+t\rangle \cdot \langle 1,4,5\rangle \, dt$$

$$= \int_0^1 4t + 20t + (10+5t) \, dt$$

$$= \cdots = \frac{49}{2} \quad //$$

Another way to write the same thing:

$$\int_C \overbrace{\vec{F}}^{\langle P,Q\rangle} \cdot \overbrace{d\vec{r}}^{\langle dx,dy\rangle} =$$

$$= \int_a^b \langle P, Q\rangle \cdot \overbrace{\left\langle \frac{dx}{dt}, \frac{dy}{dt} \right\rangle}^{\frac{d\vec{r}}{dt}} dt$$

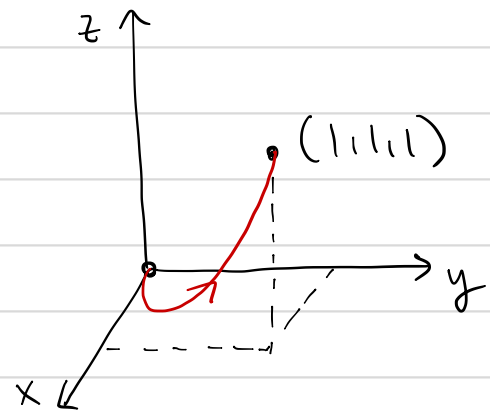
$$= \int_a^b \left(P \frac{dx}{dt} + Q \frac{dy}{dt} \right) dt$$

$$=: \int_C P\,dx + Q\,dy$$

Example:

C = twisted cubic curve

$$\begin{cases} x = t \quad y = t^2 \quad z = t^3 \\ 0 \le t \le 1 \end{cases}$$



$$\vec{F} = \langle \overbrace{xy}^{P}, \overbrace{yz}^{Q}, \overbrace{zx}^{R} \rangle$$

$$\int_C \vec{F} \cdot d\vec{r} =$$

$$= \int_C xy\,dx + yz\,dy + zx\,dz$$

$$= \int_0^1 t^3 \cdot dt + t^5 \cdot 2t\,dt + t^4 \cdot 3t^2\,dt$$

$$= \int_0^1 t^3 + 5t^6 \, dt$$

$$= \frac{t^4}{4} + 5\frac{t^7}{7} \Big|_0^1$$

$$= \frac{1}{4} + \frac{5}{7} = \frac{7+20}{28} = \frac{27}{28} \quad //$$

Work = Change in kinetic energy

$\vec{r}(t)$ = position of particle at time t.

$\vec{F}(t)$ = net force on particle at time t.

$$= m \cdot \vec{a}$$
$$= m \cdot \vec{r}''$$

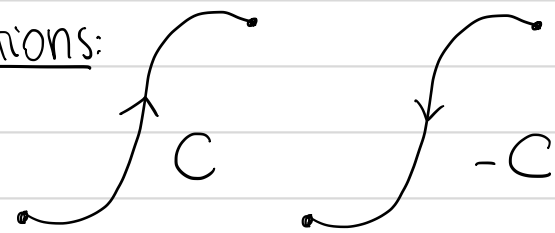
$$K = \frac{1}{2} m v^2 \quad \text{(Kinetic energy)}$$
$$= \frac{1}{2} m \vec{r}' \cdot \vec{r}'$$

$$\frac{dK}{dt} = \frac{1}{2} m \left(\vec{r}'' \cdot \vec{r}' + \vec{r}' \cdot \vec{r}'' \right)$$
$$= m \vec{r}'' \cdot \vec{r}'$$
$$= \vec{F}(t) \cdot \vec{r}'(t)$$

Conclusion:

$$\Delta K = \int_a^b \frac{dK}{dt} \, dt$$
$$= \int_a^b \vec{F} \cdot \frac{d\vec{r}}{dt} \, dt$$
$$= \int_C \vec{F} \cdot d\vec{r} = \text{Work}$$

Orientations:



1) $\int_{-C} \vec{F} \cdot d\vec{r} = -\int_C \vec{F} \cdot d\vec{r}$

2) $\int_{-C} f \, ds = +\int_C f \, ds$